Micah Shaw

Curriculum Vitae

Hong Kong — godcoderxyz@gmail.com — github.com/godCoderXYZ

Education

Victoria Shanghai Academy *2024-26 — Age 15-17*
Hong Kong
High School, IBDP
Mathematics AAHL, CS HL, Physics HL, Economics SL, English L&L SL, French ab initio SL
IBDP Score: 44/45

Carnegie Mellon University *2025 Jun-Aug — Age 16*
Pennsylvania, USA
Pre-College Summer Session
15-122 Principles of Imperative Computation (Grade: A)
Amortized Analysis, Hash, AVL Trees, Graphs and MSTs, VMs, Union Find, Priority Queues
21-127 Concepts of Mathematics (Grade: A)
Proof, induction, functions, sets, infinite sets, relations, combinatorics, number theory, modular arithmetic

Purdue University *2026 Fall — Age 17-Present*
Indiana, USA
Undergraduate CS and Math double major

Projects and Competitions

CodingQuest Participant *2024 — Age 15*
Organizer: Paul Baumgarten
Individual Ranking: 14/2144 globally
Certificate — More Info
Learned Dynamic Programming, Python File I/O and List Comprehension
Contest Programming

Developed PassPersona *2023-24 — Age 15*
Project Name: PassPersona
Github Repo
Pygame, Speech Diarization, Voice Recognition, TTS, Hashing Techniques, Encryption, PostgreSQL, Multi-threading
Voice-controlled AI Password Management Solution

Founded NextWave Hackers *2024 — Age 15*
Institution: Victoria Shanghai Academy
35 members in 2024-25 — 29 members in 2025-26
School Computer Science Club
Hosted 6 (2024-25) and 4 (2025-26) Innovation Week Events, Esports Brawl Stars Interhouse Competition, Game Theory Club Integration Seminars

Developed Game Theory Simulation *2024-25 — Age 16*
Project Name: Game Theory Simulation
Website Link—Github Repo
ReactJS, Websockets (socket.io library), Client-Server Architecture
Cross-Device Multiplayer Web Game, teaching tool (Prisoner's Dillema, Battle of the Sexes, Chicken, etc)

JA Code for Impact *2024-25 — Age 16*
Institution: Junior Achievement (JA)
Project Name: FoodLink
2nd Runner Up in Hong Kong
Github Repo
React Native, Image Classification, VQA (Visual Question Answering), MLM (Masked Language Modelling)
Food Inventory and Waste Management Solution

Counseling Website *2024-26 — Age 16-17*

Institution: Victoria Shanghai Academy
Project Name: Counseling Website
Website Link—Github Repo
NextJS Frontend, DRF (Django Rest Framework) Backend, PostgreSQL, Clerk Authentication, LangChain RAG Agent, Tool-Calling, Web Scraping, Webhooks, Virtual Memory, Multithreading, Async, Tailwind CSS, API Proxy
Counseling Website for School Social-Emotional Counseling Team

Found Optimal Strategy for Triple Down Poker *2026 — Age 17*
Startup: Sage LLC
Project Name: Triple Down Poker, a patent-pending casino table poker variant
Team of 5
Value-based RL model trained via self-play, Explainable AI (XAI), Combinatorics and Dimensionality Reduction, Expected Value and Bernoulli Random Variables
Collaborated on github and presented findings with a simplified evidence-based human strategy.

Events

Hosted Game Dev: Roblox Studio *2024 — Age 16*
Institution: Victoria Shanghai Academy
Internal Workshop-Style Seminars—28 students (13-15 yrs) registered
Obstacle Course (Obby) Development with self-made Imagination2Obby module, Roblox AI-Assistant Beta Feature

Lead Organizer of Esports Brawl Stars Interhouse Competition *2025 — Age 16*
Institution: Victoria Shanghai Academy
Internal Interhouse Competition—4 student leaders, 51 students (11-18 yrs) registered

Hosted Deep Dive into AI! *2025 — Age 17*
Institution: Victoria Shanghai Academy
Internal Workshop-Style Seminars—21 students (15-17 yrs) registered
Agentic n8n AI workflow development, NNs through MNIST digit recognition case study with self-made digit recognizer builder website

Co-Hosted Game Theory x NextWave Hackers Seminars *2025 — Age 17*
Institution: Victoria Shanghai Academy
2 hosts, $\sim$ 50 participants
Nash Equilibria, Rationality, Common Knowledge of Rationality, K-level Reasoning, Strategy Domination, Game Theory Simulation Web Game

Student Talk: Exponentials and Differential Periodicity *2026 — Age 17*
Institution: Purdue University, Math Club
$\sim$ 50 participants
Recurrence relations, modular arithmetic and lcm, augumented matricies and solvability of linear systems of equations